HIGH PERFORMANCE THERMAL INTERFACE MATERIAL FOR ENERGY STORAGE DEVICES AND OTHER ELECTRONIC COMPONENTS

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ABSTRACT

To regulate the temperature of an energy storage system for Navy shipboard use, a thermal interface material must be chosen to adhere the battery cells to the surrounding heat sink structure that effectively facilitates heat flow while withstanding harsh service conditions and adheres to other necessary performance requirements. The suggested material is a composite with aligned multiwall carbon nanotubes dispersed within a polycarbonate matrix. This combination allows for all the easy handling and servicing, gap-filling, shock resistance and chemical stability of a polymer with the thermal conductance of the carbon nanotubes. Many potential and reliable processing methods are being developed for this composite and the available testing methods are sufficient to validate the material adherence to its performance specifications and functional requirements.

NOMENCLATURE

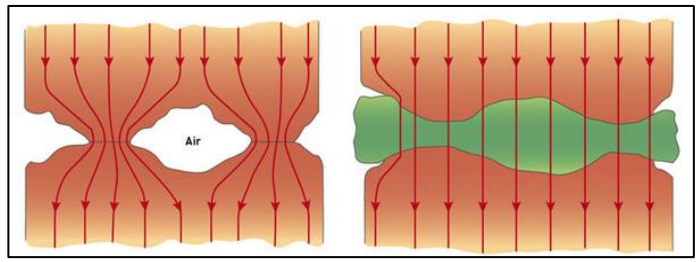
TIM	Thermal Interface Material
ASTM	American Society for Testing and Materials
SOA	State-of-the-Art
PCM[1]	Phase Change Material
PCM[2]	Polymer Composite Matrix
CNT	Carbon Nanotubes
MWCNT	Multiwall Carbon Nan
PC	Polycarbonate

PRIMARY OBJECTIVE

The technical objective of this topic is to develop a thermal interface material for use in militarized battery modules, which has the following characteristics: robust to vibration and abrasion, non-permanent bonding, high dielectric strength, and high thermal conductance.

PROBLEM SIGNIFICANCE

Energy storage systems are imperative for the United States' Navy, especially for shipboard applications where energy is needed in isolated environments. Energy storage systems incorporate battery packs with individual battery modules and cells and during operation, the system emits heat. If the heat is not dissipated properly to regulate the temperature within operating parameters of the system, it can affect the performance; reduce the cycle life; compromise safety; and cause an unbalanced battery system. Systems are often placed in a heat sink structure to conduct the heat from the system to the fins to dissipate the heat more effectively through convection. For proper conduction to the fins, the surface of the battery module must be connected to the heat sink. The surfaces, however, contain imperfections and when they are simply put into contact with one another, air voids are present between the uneven surfaces that behave as insulators and force the heat to transfer through the high point of contact, as shown in Fig. (1a). This slows the rate of heat transfer and can allow the energy storage system to overheat, producing the aforementioned issues.



(A) **(B)**

FIGURE 1. SURFACE CONNECTIONS (A) INTERSTICIAL AIR VOIDS (B) THERMAL INTERFACE MATERIAL

To improve the flow of heat away from the system, a thermal interface material is used in between the surfaces that fills the air voids and facilitates heat conduction, shown in Fig (1b). Materials with a low thermal impedance are optimal for this application and other properties of the material can be obtained by manipulating the composition and elements in the material to better suit the application. There is a large variety of thermal interface materials, however not all are suitable for applications in militarized battery modules.

Energy storage systems for Navy shipboard use shall be operated at a high rate and high duty cycle, leading to a significant thermal management challenge. The thermal interface material chosen for this application must provide optimal performance while given the rigorous and specific service conditions to effectively manage the system temperature.

BACKGROUND

Material Requirements

The material must meet the following requirements to effectively manage the temperature of the system for applications in militarized battery modules. Some considerations when selecting a material are that the material must cover a heat transfer surface area. The surface area for a single battery cell is 8 sq.in. to 60 sq.in. and the number of cells per module is 12 to 48 and therefore, the material must be able to cover a large surface area. The TIM will be applied to machined aluminum surfaces and the gap width across the interface is expected to be approximately 50mil with 100psi pressure, although larger gap sizes should be permitted if desired and the material must be able to fill these voids. The heat flux at the cell surface must be between 3 kW/m^2 to 7 kW/m^2 and the expected operating

range for the TIM is between 40°C and 70°C with a maximum temperature of 150°C. Thermal conductance must be at least 2000 W/m^2K, however a conductance of 5000 W/m^2K would be ideal for optimum heat transfer. The electrical insulation rating for the material must have a threshold of 2000V but the objective is 5000V. Compliance with recognized standards for plastic materials including ASTM D1000 and UL 94 should be met, with corresponding standards met for other material types. The material must comply with MIL-S-901D for mechanical shock resistance and MIL-STD-810G for transportability and other environmental compatibility. Table (1) shows the compliance matrix for the TIM.

TABLE 1. COMPLIANCE MATRIX

Item No.	Feature to be Tested	Pass/ Fail Criteria
1	Area of Coverage (single cell)	≥ 60 sq.in.
2	Heat Flux at Cell Surface	3kW/m² - 7kW/m²
3	Adherence to Surface	Machined Aluminum
4	Operating Temp	40°C - 70°C
5	Max Temperature	150 °C
6	Gap Coverage	≥ 50 mil (@100psi)
7	Thermal Conductance	2000 W/m²K - 5000 W/m²K
8	Electrical Insulation Rating	2000 V - 5000 V
9	Flammability	ASTM D 1000, UL 94
10	Mechanical Shock Resistance	MIL-S_901D
11	Vibration Resistance	MIL-STD-167-1A
12	Transportability/ Environmental Compatibility	MIL-STD-810G
13	Bonding	Non-Permanent

Criterion for Selecting Materials

The thermal interface material (TIM) must be designed for use in militarized battery modules. The material shall be thermally conducting and electrically insulating and withstand vibration and abrasion. To allow for ease of servicing, the TIM must not form a permanent bond between the surfaces. Either cylindrical or prismatic large format battery cells may be used, of which produce varying gap widths between the surfaces, therefore the thermal interface material must accommodate the irregular gap width and clamping force. To ensure safety of the personnel, the TIM must be non-flammable, non-toxic and not require the use of safety equipment for handling. Table (2) below displays the various criteria that shall be used during consideration and assessment of the materials. Each criteria is weighted and the materials shall be evaluated with a 1, 0 or -1 depending on their adherence to the property. The overall scores shall be evaluated and the TIM with the highest score shall be chosen.

TABLE 2. CRITERION SELECTION MATRIX

Item No.	Criteria	Weight	Potential Materials			
			I	II	III	IV
1	Thermal Conductivity	20				
2	Electrical Insulation	15				
3	Vibration and Abrasion Resistant	10				
4	Non-Permanent Bonding	20				
5	Accommodate Varying Gap Width	15				
6	Safety	20				

Preference is given to advanced materials (e.g. composites, phase change materials and nanomaterials) with unmatched thermal performance when considering the aforementioned challenges.

Currently Used Materials

Current state-of-the-art (SOA) materials may provide a subset of the aforementioned traits. For example, gap pads may be used in irregular gaps but their thermal conductance is poor compared to epoxy-type materials. While epoxy-type materials have high thermal conductance, they form a semi-permanent or permanent bond, which hinders the disassembly and servicing of the battery module. Thermal grease materials may leak from the module over time, and may have issues maintaining electrical insulation between the cell and heat sink. Finally, many current SOA materials are unproven in a militarized environment subject to shock, vibration, transportability, and handling requirements.

Thermal Pads are a family of soft, thermally conductive silicon and non-silicone elastomers for applications where heat must be conducted over a large and variant gap. They are used to bridge large gaps without stressing components beyond their safe limits. They can include carrier (fiberglass) for added strength. A super thermal gap filler interface material has a silicone base and excellent thermal conductivity reaching as high as 13.2 W/mk. With operating temperatures from -40°C to 150°C, it falls well within specifications. These gap fillers are spongy and in a solid state during operation. They have a low thermal resistance of 1.1°Ccm^2/W. There is the option of having an adhesive applied to one side so it sticks to something having it without. They have puncture, shear, and tear resistance so they have the ability to stand up to rough handling over time as well as the ability to contour to irregular surface finishes.

Thermal Gels are supplied as single component compounds that can be dispensed over the heat generating component. Lower mechanical stress on components, provides low thermal impedance at thin and thick gaps, allowing use of common heat spreaders. They have proven reliability in extreme temperature cycling and stock and vibration.

Thermal Greases have excellent thermal conductivity and the viscosity lowers as temperature is increased thus becoming more effective. Grease is placed between two contact surfaces to eliminate air pockets and increase thermal conductivity. Thermal grease is a very popular application for many of its properties. It has a polymerized liquid matrix and with its silicon base is very thermally conductive and electrically insulating.

Phase Change Materials (PCM[1]) are designed to maximize heat sink performance and improve component reliability. At

room temperature, they are solid allowing easily handling and installation, at operating temperatures the material melts and fills gaps. Paraffin composite phase change materials has many benefits and can accomplish what has been asked. It has the ability to absorb and retain latent heat from an operation and store it safely as it is relieved to a heat exchanger. This phase change material or PCM[1] does however have a low thermal conductivity. It is supposed to absorb the heat and transfer it later rather than transfer it all in a short amount of time. The paraffin wax has a density change of 0.057 kg/L as it goes from liquid to solid as well as a change in specific heat which is not desirable for a military application.

Polymer Composite Matrix (PCM[2]) are composites comprising of a polymer matrix with particles of another form dispersed inside the matrix to enhance or combine properties of multiple materials. Polymers are desirable for their electrical insulation, easy handling and machining, wide range of strength and abrasion resistance, and corrosion resistance. Its major setback is its low operating temperature and poor thermal conductance. Nanoparticles with more desirable properties can be added to the polymer matrix to increase its thermal conductivity while maintaining its more desirable polymer properties.

MATERIALS SYSTEMS PROPOSED

The material best suited for applications in militarized battery modules is a polymer composite matrix material. The polymer shall be made of polycarbonate and aligned carbon nanotubes (CNT) shall be dispersed throughout the matrix. The polycarbonate is a durable thermoplastic with a high impact-resistance. A low molecular mass grade is easier to mold and shall form better to the gap irregularities present. Polycarbonate is a good electrical and thermal insulator. To enhance the thermal capabilities of the polycarbonate, aligned carbon nanotubes shall be dispersed throughout the matrix with the ends of the nanotubes in contact with the heat transfer surfaces for efficient heat transfer. Carbon nanotubes are nanostructures with a high thermal conductivity and are extremely flexible. Combining these materials yields the best combined properties that shall meet the requirement necessary for this application than any single material. The final composite shall be a pre-cut plastic piece that is placed in the desirable spot. The low grade plastic shall form to fill gaps when the operating temperatures are reached, it may be necessary to raise the temperature above the range to ensure gaps are filled. The carbon nanotube ends will come into contact with the surfaces allowing the heat transfer from the battery cell the heat sink. For servicing, the system can be shut off and allowed to cool and the material can simply be gently scraped off, and should hold its form.

MATERIALS PROCESSING CHALLENGES AND SOLUTIONS

The chosen polycarbonate/aligned carbon nanotubes composite fits all requirements and has been proven through research to be an excellent thermal interface material. Although this is the ideal material, its preparation process can be extremely complex. This material can be prepared in many different ways and because this composite has many desired properties, new innovative methods are constantly being researched.

A process that has been shown to produce a quality composite is through chemical vapor deposition and twin-screw compounding extruder method. Although this method of processing has only recently been researched, it has been evaluated as a method which produces great dispersion as well as alignment and interaction between the polycarbonate and carbon nanotubes. First, the MWCNT are synthesized from methane through catalytic chemical vapor deposition. This processing approach involves reduction from its oxidized state in a flow of hydrogen followed by a temperature manipulation from 1073 K down to 973 K, then finally lowered to ambient temperature under a flow of nitrogen. The nanotubes are then treated with an acid to remove any remaining catalyst particles. The polycarbonate starts out in the form of granules and is melt-compounded with the MWCNT using a twin-screw compounding extruder. This extruder, such as the one seen below in Figure (2), mimics mixing behavior utilizing large, co-rotating, twin-screw compounding machines, while offering precise control of mixing temperature and speed [4].

FIGURE 2. TWIN-SCREW COMPOUNDING EXTRUDER

Several issues that arise from this method of processing are in regard to purity, dispersion and defects. Purity of MWCNTs are a concern due to the residual metallic catalysts produced during processing which can be detrimental to performance. In addition, the quality of dispersion will need to be a consideration as the dispersion of nanotubes will affect the performance of the material. These issues may be prevented through observing the nanotubes using scanning electron microscopy. Defects often depend on the number of layers within the multiwall carbon nanotube. Thermogravimetric analysis can be used to obtain information about uniformity and ensure defects are minimal to non-existent. It is further recommended that the MWCNTs and the PC/MWCNT composite be evaluated using Raman spectroscopic analysis. These results are interpreted through analyzing the changes in Raman shift. This analysis further investigates the dispersion and interaction between the polycarbonate and nanotubes [5].

Although this material requires this tedious processing method and involves many possibilities for error, the produced material is ideal and precautionary measures may be taken to produce a quality thermal interface material.

PROPOSED EVALUATION METHODS

Evaluating the performance of the chosen materials system is a vital part of ensuring the correct choice has been made. While knowledge of material properties and consideration of past research may be utilized to gather an assumption that the decided system will work, further evaluation in the form of

testing will validate this decision. A standard test method for thermal interface materials may be found in ASTM D5470. This standard is a test method for thermal transmission properties of thermally conductive electrical insulation materials [2]. By measuring the steady state thermal impedance of the material in question, this standard evaluates the thermal conductivity and thus assists in determining the validity of polycarbonate/aligned carbon nanotube matrix as a TIM. Shown below in Fig. (3) is an apparatus useful to implement the aforementioned standard ASTM D5470.

FIGURE 3. THERMAL INTERFACE MATERIALTESTER

This Analysis Tech TIM Tester is an automated test system that is capable of measuring the thermal resistance and conductivity of thermally conductive materials. Materials most commonly testing are the type frequently used for electronic packaging such as the application discussed. The specimen is first clamped between one heated and one cooled surface; this setup ensures a heat flow through the test sample. The thermal conductivity is found through the temperature gradient and the induced heat-flow. The effect of contact pressure on the thermal performance of the material is also evaluated through pressure-batch mode testing [4]. In addition to testing the desired properties of the material by itself, it is also important to do preliminary testing of the thermal interface material implemented within the militarized battery module. This testing will allow further confidence that the polycarbonate/aligned carbon nanotubes matrix will fulfill all requirements and maintain the battery's life and function. For this purpose, evaluating the temperature of the heat sink and the battery itself against time will be necessary; placement of temperature sensors may be used to accomplish this task. Additionally, the ability to withstand vibration and abrasion is important for this application as it will be used aboard navy ships. To test the capability of the battery module with implemented TIM, a vibration test will be performed using a vibration testing machine in which induced shock will occur over a high frequency. Abrasion resistance will be tested using taber abrasion testing following all standards. This process involves mounting the flat specimen to a turntable and rotating it between two abrasive wheels. The specimen mass is used to allow for material property comparison [6]. Utilizing these testing procedures will ensure the material proposed meets all desired specifications and performs the desired task.

CONCLUSIONS

The proposed material, incorporating multiwall carbon nanotubes dispersed in a polycarbonate matrix, is a strong contender for applications in militarized battery modules for navy shipboard use. The composite combines the desirable properties of shock resistance, chemical stability, easy handling and electrical insulation of the polycarbonate with the high thermal conductivity of the CNT. The processing methods are still under development, however chemical vapor deposition in conjunction with twin-screw compounding extruder has shown to produce a reliable composite. Extensive testing is necessary to validate the properties of the material for adherence to the requirements, but all is testing methods are available to do so. The composite has been tested for thermal conductivity and has produced promising results; and with knowledge of the behavior of the polycarbonate, the composite shows promise to meet the remaining requirements.

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